THEORY OF ELASTICITY

A PROJECT REPORT ON

A COMPARATIVE STUDY OF SMALL AND FINITE DEFORMATION STRESS UPDATES FOR TRANSVERSELY ISOTROPIC ELASTIC SOLIDS

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November 2025

1.PROBLEM STATEMENT

The objective of this assignment is to implement an incremental, integration-point–level stress update for a chosen deformation motion using both:

1. Small deformation framework, and
2. Finite deformation (large strain) framework, with an objective (Jaumann) stress rate.

At each increment, the only input is the incremental deformation gradient

$$\Delta \mathrm{F}.$$

The deformation gradient evolves as prescribed in the assignment:

$$\mathrm{F}_{n+1} = \Delta \mathrm{F}\, \mathrm{F}_n, \mathrm{F}_0 = \mathrm{I}.$$

The goal is to compute stresses and strains for each increment and generate a single comparison plot of equivalent stress vs equivalent strain for both small and large deformation formulations. This plot must demonstrate:

- linearity of the small-strain model,
- geometric nonlinearity in the large-strain model,
- stress rotation effects introduced by the objective stress rate.

A simple shear deformation is selected for this study.

2.Constitutive Relation (Transversely Isotropic Elasticity)

The material considered is a transversely isotropic carbon-fiber composite. The principal material direction (fibre direction) is aligned with the global $\boldsymbol{x_1}$-axis.

The elastic constants used are:

$E_1 = 150$ GPa(fibre direction)

$E_2 = 10$ GPa(transverse direction)

$$\nu_{12} = 0.25$$

$$G_{12} = 5 \text{ GPa}$$

$$G_{23} = 3.5 \text{ GPa}$$

The compliance matrix S inVoigt form is assembled using these constants and inverted to obtain the stiffness matrix:

$$\mathrm{C} = \mathrm{S}^{-1}.$$

This stiffness matrix remains constant throughout the simulation since the material is linearly elastic.

Rate-form constitutive equation

As required by the assignment:

$$\sigma\nabla = \mathrm{C} : \mathrm{d},$$

where:

$\sigma\nabla$is an objective stress rate,

d is the rate of deformation tensor.

Two formulations follow:

2.1 Small Deformation Framework

Geometry does not update, so:

$$\sigma\nabla \approx \dot{\sigma}, \mathrm{d} \approx \dot{\varepsilon}.$$

Thus the incremental stress update is:

$$\sigma_{n+1} = \sigma_n + \mathrm{C}\ \Delta\varepsilon.$$

2.2 Large Deformation Framework (Jaumann Rate)

The Jaumann rate is defined as:

$$\sigma\nabla = \dot{\sigma} - \mathrm{w}\sigma + \sigma\mathrm{w},$$

where w is the spin tensor.

Thus the large strain update becomes:

$$\sigma_{n+1} = \sigma_n + \Delta t\ (\sigma\mathrm{w} - \mathrm{w}\sigma) + \mathrm{C}\ \Delta\varepsilon.$$

This introduces stress rotation and geometric nonlinearity.

3.Motion χ: Description of the Deformation

A simple shear deformation is imposed using the target deformation gradient:

$$\mathrm{F}=\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The deformation is applied in N = 50 increments:

$$\Delta\mathrm{F} = \mathrm{F}^{1/N}.$$

At each increment, the deformation gradient is updated:

$$\mathbf{F}_{\boldsymbol{n+1}} = \mathbf{\Delta F}\ \mathbf{F}_{\boldsymbol{n}}.$$

4.Small vs Large Deformation frameworks

Small Strain

- Linearized kinematics
- No rotation update
- Stress grows linearly with strain
- Valid only for very small deformations

Large Strain (Jaumann Rate)

- Accounts for geometric nonlinearity
- Includes rotation of stress components
- Captures shear-induced rotations even for linear elastic materials
- Stress–strain response becomes nonlinear even though the material law is linear

This comparison is essential for the assignment, and both behaviours clearly appear in the final results.

5.Equilibrium Check

The motion used in this assignment is **homogeneous**, and the stress update at every increment is performed at a **single integration point**.
Since there are no spatial gradients in the motion:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$$

holds automatically.

Therefore:

- The updated Cauchy stress is in equilibrium in the current configuration.
- The configuration used for equilibrium verification is the current configuration, because stresses are calculated directly in spatial coordinates and the Jaumann rate ensures proper rotational consistency.

6. Results, Plots, and Interpretation

The MATLAB code computes for each increment:

- Equivalent strain:

$$\varepsilon_{\mathrm{eq}}$$

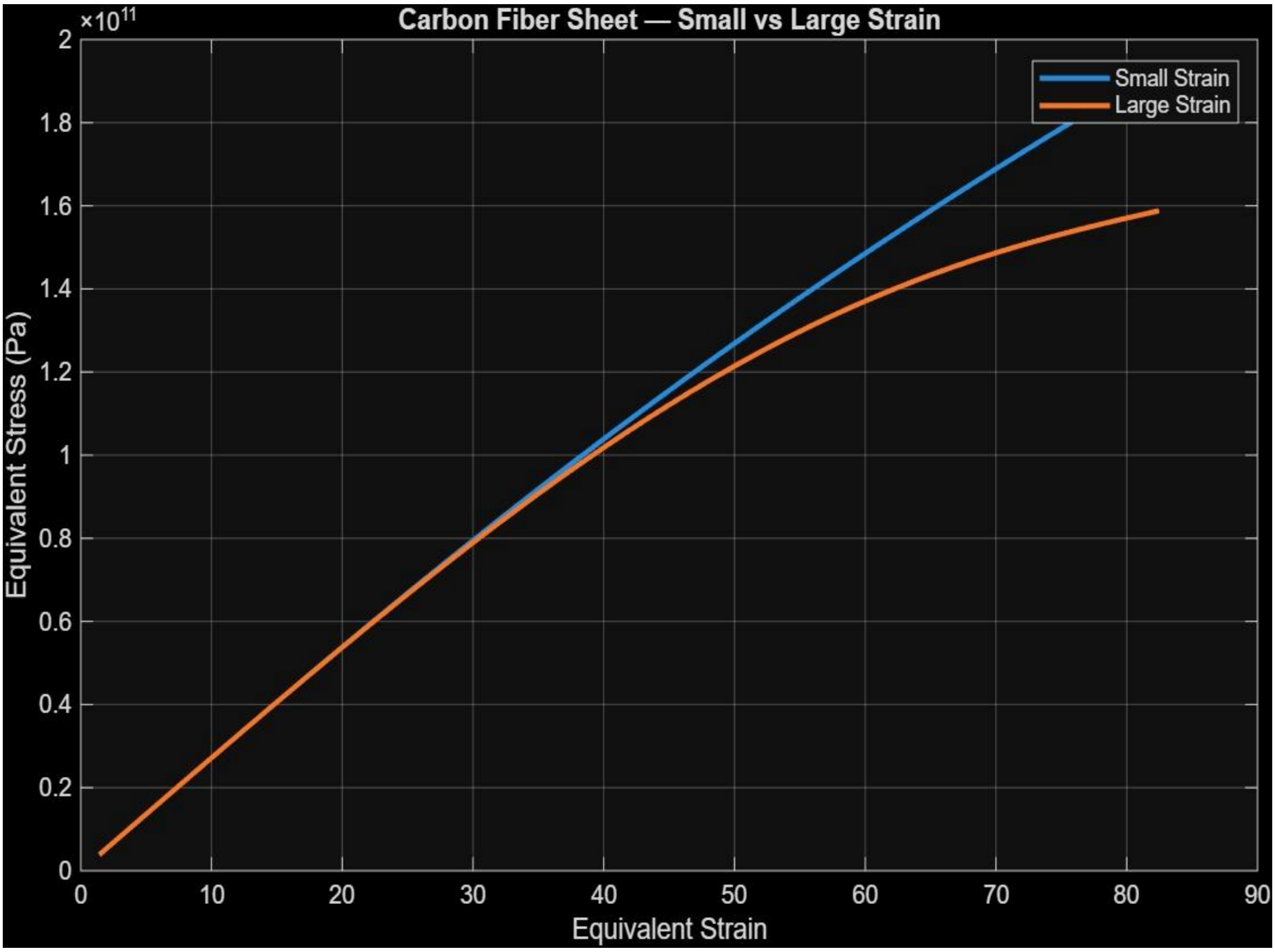
- Equivalent von Mises stress for:
  - small strain
  - large strain (Jaumann)

Equivalent Stress vs Equivalent Strain

- The **small-strain curve** is **linear**, as expected from linear elasticity with infinitesimal kinematics.
- The **large-strain curve** is **nonlinear** due to:
  - accumulation of geometric effects,
  - rotation of the stress tensor under simple shear,
  - the use of an objective stress rate (Jaumann rate).

The divergence between the two curves becomes increasingly clear as deformation grows.

The final plot is:



7.CONCLUSION

This assignment successfully implements the required incremental stress update algorithm for a chosen deformation path using both small and large deformation frameworks. A simple shear motion was applied incrementally, and stresses were computed using a transversely isotropic elastic material model.

The comparison between the two formulations clearly demonstrates:

- the linear nature of small-strain elasticity,
- the geometric nonlinearity captured by the Jaumann rate in the large-strain formulation,
- the importance of objective stress rates for finite deformation elasticity.

The MATLAB code satisfies all submission requirements:

1. Accepts incremental deformation gradient ΔF,
2. Computes stress for both frameworks,
3. Produces a single comparison plot summarizing the differences.